Formula Sheet: Electric and Magnetic Fields

Formulas

Definition of electric field :	$\vec{F_E} = q\vec{E}$	Electric field strength:	$E = \dfrac{F_E}{q}$
Electric field strength of a point charge:	$E = k\,\dfrac{Q}{r^2}$	Field of a long, straight wire:	$B = \dfrac{\mu_0}{2\pi}\,\dfrac{I}{r}$
Newton's second law:	$F = m\,a$	Weight:	$F_G = mg$

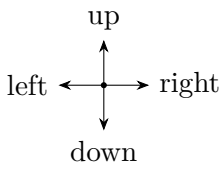
Constants

$k = \dfrac{1}{4\pi\varepsilon_0} = 8.99 \cdot 10^9\,\mathrm{N\,m^2/C^2}$ $\qquad$ $\mu_0 = 4\pi \cdot 10^{-7}\,\mathrm{T\,m/A}$

$e = 1.602 \cdot 10^{-19}\,\mathrm{C}$ $\qquad$ $m_e = 9.109 \cdot 10^{-31}\,\mathrm{kg}$

$g = 9.81\,\mathrm{m/s^2}$

Directions



All directions refer to the page held upright in front of you: *up*, *down*, *left* and *right* as drawn on the page; gravity points *down*.

⊙ out of the page (towards you) $\qquad$ ⊗ into the page (away from you)